IBM **Quantum**

OFFICIAL EVENT

Qiskit | Fall Fest | 2026

Noise, Backends & Real Hardware



Qiskit Fall Fest 2026 at University of Engineering & Technology Lahore,Narowal Campus

Review

What have we learned so far?



- **Qubits**
  - State vectors, Bra-ket notation, Basis states
- **Measurement**
  - Probability amplitudes, Born rule
- **Multiple Qubits**
  - Tensor product, Product states, Entangled states
- **Operators**
  - Unitary matrices, Hermitian matrices
- **Circuits**

Qubits

States as Vectors



Qubits are often represented with a ***column vector*** called a ***ket***. Each ket has an associated ***row vector*** called a ***bra***.

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \quad \text{and} \quad \langle\psi| = \begin{bmatrix} \alpha^* & \beta^* \end{bmatrix}$$

α and β are ***complex numbers***, and α^* and β^* are their ***complex conjugates***.

Complex conjugation means that if $\alpha = x + iy$, then $\alpha^* = x - iy$.

Importantly, α and β are ***normalized***: $|\alpha|^2 + |\beta|^2 = 1$

Qubits

Vectors Need a Basis

We always choose a ***basis*** to work in. The *standard* or *computational* ***basis states*** in quantum computing are:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

A qubit in a ***pure state*** can be written in several equivalent ways:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

This ***linear combination*** of basis states is called ***superposition***.

Measurement
Coefficients are Probability Amplitudes

By the ***Born rule***, probabilities are the squared magnitudes of the amplitudes.

The ***probability amplitudes*** for $|0\rangle$ and $|1\rangle$ are the coefficients α and β, so:

$$P(0) = |\alpha|^2 \quad \text{and} \quad P(1) = |\beta|^2$$

This is why the coefficients must be normalized.

$$|\alpha|^2 + |\beta|^2 = 1$$

Note, if $\alpha = x + iy$ then:

$$|\alpha|^2 = \alpha^*\alpha = (x - iy)(x + iy) = x^2 + y^2$$

Measurement
Finding an Outcome

Measuring in the computational basis gives either 0 or 1.

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$$

Applying the Born rule to $|\psi\rangle$, the probabilities of measuring either 0 or 1 are:

$$P(0) = \left|\frac{\sqrt{3}}{2}\right|^2 = \frac{3}{4} \qquad P(1) = \left|\frac{1}{2}\right|^2 = \frac{1}{4}$$

After measurement, the qubit's state becomes the basis state of the measured outcome.

$$0 \to |0\rangle \qquad 1 \to |1\rangle$$

Multiple Qubits

Two Qubits, Four Basis States

Two qubits have four associated basis states:

$$|00\rangle,\ |01\rangle,\ |10\rangle,\ |11\rangle$$

The associated state vector has four terms:

$$|\psi\rangle = a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle$$

Amplitudes determine the probabilities of each result, so they must be normalized:

$$|a|^2 + |b|^2 + |c|^2 + |d|^2 = 1$$

Multiple Qubits
Combining States

The ***tensor product*** (also known as the ***Kronecker product***) combines the states of multiple qubits.

$$|0\rangle \otimes |1\rangle = |01\rangle \qquad |1\rangle \otimes |0\rangle = |10\rangle$$

Generally:

$$\begin{bmatrix} a \\ b \end{bmatrix} \otimes \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} ac \\ ad \\ bc \\ bd \end{bmatrix}$$

Multiple Qubits

States We Can Separate

If a multi-qubit state can be written as a tensor product of individual pure states it is called a ***product state***.

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad \rightarrow \quad |+\rangle \otimes |+\rangle = \frac{1}{2}\Big(|00\rangle + |01\rangle + |10\rangle + |11\rangle \Big)$$

However, some states cannot be written as a product state, for example:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Multiple Qubits

States We Cannot Separate

When a state cannot be written as a tensor product of individual pure states it is called an ***entangled state***.

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

A measurement of either qubit determines the outcome of a measurement of the other qubit.

For $|\Phi^+\rangle$, the only possible outcomes for a measurement in the computational basis are $|00\rangle$ and $|11\rangle$. Each with probability $1/2$.

Operators

Matrices Act on States

Operators are linear transformations that act on the vectors representing quantum states. They are represented by matrices:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

If an operator A acts on $|\psi\rangle$, then: $A|\psi\rangle = |\psi'\rangle$

$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{and} \quad X|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Operators
Evolution and Measurement

Unitary operators describe reversible evolutions of quantum states.

$$U^\dagger U = UU^\dagger = I$$

Hermitian operators describe ***observables***, their real eigenvalues are the possible measurement outcomes for that observable.

$$A^\dagger = A$$

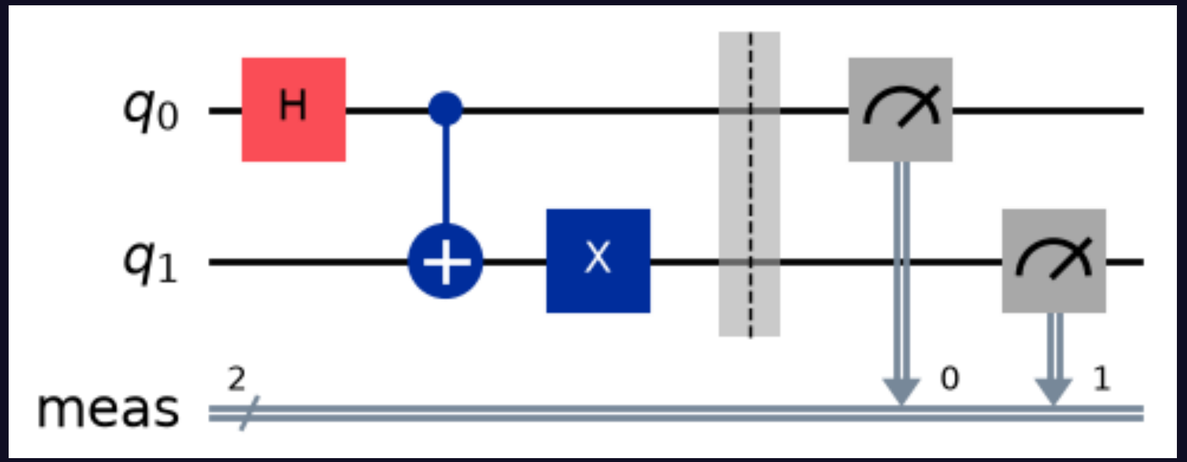
Circuits
Wires and Gates

It is common to represent qubits and operators as a ***circuit***.

Qubits are represented as wires, with each gate acting on them from the left to the right.

Single qubit gates: X, Y, Z, H

Multi-qubit gates: $CNOT, SWAP$



Topics

- **Noise in Quantum Computers**
  - Where Does Noise Come From?
- **Describing Noise**
  - State Vectors Are Not Enough, Interpreting a Density Matrix, Why Density Matrices Matter, Mixed States, Operators on Density Matrices
- **Noise Models**
  - Representing an Imperfect Operation, Randomizing the State, Depolarization in Action, Random Gates, Pauli Noise in Action, Relaxation and Decoherence, Energy Relaxation, Phase Coherence, Thermal Relaxation, Measuring T_1 and T_2
- **Simulating Noise**
  - Ideal vs, Noisy Simulation, Simulating From a Backend, Building a Custom Noise Model

Noise in Quantum Computers

Where Does Noise Come From?

Real qubits are physical systems.

Qubits interact with their environment, the QPU's control operations are imperfect, and the final measurements can be incorrect.

These are all sources of noise:

- ***Decoherence*** accumulates as time passes.
- ***Gate errors*** are caused by imperfections in the operations applied.
- ***Readout errors*** happen when the measured result is incorrect.

Describing Noise

State Vectors Are Not Enough

Until now we have represented pure states as state vectors. For a single qubit:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

States can also be more generally represented as ***density matrices***:

$$\rho = |\psi\rangle\langle\psi| = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \begin{bmatrix} \alpha^* & \beta^* \end{bmatrix} = \begin{bmatrix} \alpha\alpha^* & \alpha\beta^* \\ \beta\alpha^* & \beta\beta^* \end{bmatrix}$$

Describing Noise

Interpreting a Density Matrix

For a single qubit: $$\rho = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \beta\alpha^* & |\beta|^2 \end{bmatrix}$$

The ***diagonal elements*** are the probabilities $P(0)$ and $P(1)$.

The ***off-diagonal elements*** describe the ***coherence*** between the basis states.

Valid density matrices are ***positive semi-definite*** and have the properties:

$$\mathrm{Tr}(\rho) = 1 \qquad \rho = \rho^\dagger$$

Describing Noise

Why Density Matrices Matter

Weighted combinations of density matrices are also density matrices when the weights are probabilities.

$$\rho = p_1\rho_1 + p_2\rho_2 + \dots$$

The state is ρ_1 with probability p_1, it is state ρ_2 with probability p_2, etc...

Probabilities still need to be non-negative and add up to one.

$$p_i \geq 0, \qquad \sum_i p_i = 1$$

Describing Noise

Mixed States



What if our state had a 50% chance of being $|0\rangle$ or $|1\rangle$?

$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}$$

By measuring in the computational basis can we differentiate it from $|+\rangle$?

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad \rho_+ = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

These states cannot be distinguished with only a Z-basis measurement.

Describing Noise
Operators on Density Matrices

Unitary operators evolving state vectors is represented by: $U|\psi\rangle = |\psi'\rangle$

If $|\psi'\rangle = U|\psi\rangle$, then $\langle\psi'| = \langle\psi|U^\dagger$ and we can write the expression:

$$|\psi'\rangle\langle\psi'| = U|\psi\rangle\langle\psi|U^\dagger$$

So the same unitary operator evolving a density matrix is written as:

$$U\rho U^\dagger = \rho'$$

Noise Models

Representing an Imperfect Operation

A ***noise model*** gives a mathematical rule for approximating how a type of noise changes the state.

The model is *NOT* a description of the exact physical process causing the noise.

The model *IS* a mathematical tool that replicates the statistics of the physical process causing the noise.

Noise Models
Randomizing the State

The ***maximally mixed state*** is the state of randomness represented by: $\frac{I}{2}$

Depolarizing noise replaces some fraction of the state with the completely mixed state. For a single qubit this can be represented as:

$$\rho \rightarrow (1-\lambda)\rho + \lambda\frac{I}{2}$$

The term λ is in the range $0 \leq \lambda \leq 1$, where $\lambda = 0$ represents no noise and $\lambda = 1$ will completely depolarize ρ.

Noise Models
Depolarization in Action

A qubit in the state $|0\rangle$ undergoes depolarization. Remember that:

$$|0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad \frac{I}{2} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}$$

If $\lambda = 1/4$:

$$\rho' = \frac{3}{4}\rho + \frac{1}{4}\frac{I}{2} = \frac{3}{4}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{1}{4}\begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} = \begin{bmatrix} 7/8 & 0 \\ 0 & 1/8 \end{bmatrix}$$

Noise Models
Random Gates

Pauli noise models the result of errors as X, Y, and Z operations which occur with specified probabilities.

Pauli noise can model random ***bit flips*** (X), ***phase flips*** (Z), or a combined ***bit-and-phase flip*** (Y) occurring during computation.

$$\rho \rightarrow p_I \rho + p_X X\rho X^\dagger + p_Y Y\rho Y^\dagger + p_Z Z\rho Z^\dagger$$

The probabilities must still be non-negative and their sum must be one.

Noise Models

Pauli Noise in Action

Starting with $\rho = |0\rangle\langle 0|$ and supposing all four probabilities are $1/4$.

$$\rho = Z\rho Z^\dagger = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad X\rho X^\dagger = Y\rho Y^\dagger = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

This gives:

$$\rho' = \frac{1}{2}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{1}{4}\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} + \frac{1}{4}\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}$$

Equal probability Pauli noise has completely randomized the state.

Noise Models

Relaxation and Decoherence



Two parameters are used to describe time dependent noise: T_1 and T_2

T_1 is the ***energy relaxation time***. It describes the time for an excited qubit to decay from $|1\rangle$ to $|0\rangle$.

T_2 is the ***phase coherence time***. It describes how quickly coherence is lost between $|0\rangle$ and $|1\rangle$.

Each parameter affects different parts of the density matrix.

Noise Models
Energy Relaxation



For a single qubit density matrix $\rho = \begin{bmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{bmatrix}$,

the probability of finding the qubit in an excited-state, given by ρ_{11}, decreases as:

$$\rho_{11}(t) = e^{-t/T_1}\rho_{11}$$

The lost probability moves to the ground state, ρ_{00} :

$$\rho_{00}(t) = \rho_{00} + (1 - e^{-t/T_1})\rho_{11}$$

Larger T_1 values mean the excited-state probability decays more slowly.

Noise Models
Phase Coherence

For a single qubit density matrix $\rho = \begin{bmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{bmatrix}$,

the coherence between $|0\rangle$ and $|1\rangle$, given by the off-diagonal terms, decreases as:

$$\rho_{01}(t) = e^{-t/T_2}\rho_{01} \qquad \rho_{10}(t) = e^{-t/T_2}\rho_{10}$$

Larger T_2 values mean coherence is retained for longer.

Noise Models
Thermal Relaxation



Putting these effects together gives us a model of how the density matrix changes over time:

$$\rho(t) = \begin{bmatrix} \rho_{00} + (1 - e^{-t/T_1})\rho_{11} & e^{-t/T_2}\rho_{01} \\ e^{-t/T_2}\rho_{10} & e^{-t/T_1}\rho_{11} \end{bmatrix}$$

Noise Models

Measuring T_1 and T_2



The values of T_1 and T_2 are derived experimentally for each qubit:

$$\text{Prepare } |1\rangle \rightarrow \text{Wait} \rightarrow \text{Measure in } Z\text{-basis} \quad (T_1)$$

$$\text{Prepare } |+\rangle \rightarrow \text{Wait} \rightarrow \text{Measure in } X\text{-basis} \quad (T_2)$$

This experiment is done for many shots at different wait durations, and the resulting statistics are used to determine the T_1 and T_2 times for each qubit.

Simulating Noise
Ideal vs. Noisy Simulation



An ideal simulation assumes perfect qubits and gates.

Qiskit Aer is a high-performance simulator that supports noise models.

Two common ways to simulate noise in Aer are to ***use a custom noise model*** or to ***use calibration data from a real backend***:

```
from qiskit_aer import AerSimulator

ideal_sim   = AerSimulator()
noisy_sim   = AerSimulator(noise_model=noise_model)
backend_sim = AerSimulator.from_backend(backend)
```

Simulating Noise
Simulating From a Backend



Aer can use the properties of a real or fake backend:

```
backend_sim = AerSimulator.from_backend(backend)
```

The backend information includes properties like the QPU's basis gates, qubit coupling map, and noise properties derived from device calibration data.

This can be used to preview how a circuit should behave on real hardware.

Using fake backends we can work locally, without an internet connection.

Simulating Noise
Building a Custom Noise Model

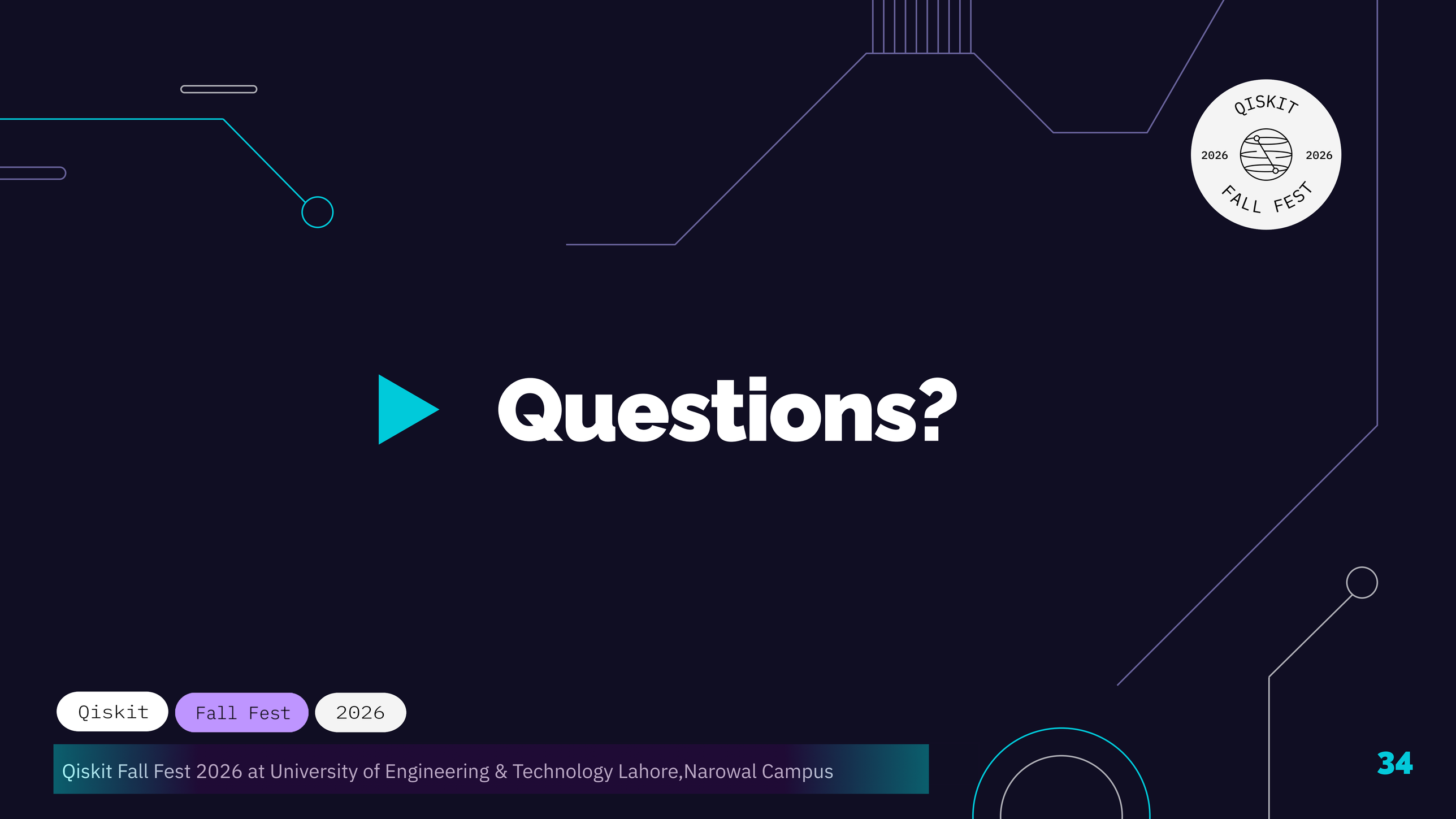
The qiskit_aer.noise package provides NoiseModel and functions to define each of the noise models we have already discussed:

depolarizing_error(param, num_qubits)

pauli_error(noise_ops)

thermal_relaxation_error(t1, t2, time, excited_state_population=0)

```
depolarizing_error(0.03, 1)
pauli_error([('I',0.25),('X',0.25),('Y',0.25),('Z',0.25)])
thermal_relaxation_error(60_000, 40_000, 100_000)
```



Questions?

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